Marco Lira Franco

Junior Data Scientist

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Summary

Junior Data Scientist with a background in Applied Mathematics and a Master's degree in Astrophysics. I develop data science projects using Python and SQL, combining statistical analysis and machine learning to solve classification, regression, time series, natural language processing, and computer vision problems. My mathematical background and experience in scientific research strengthen my ability to model problems, interpret results, and turn data into useful insights for business and analytical decisions.

Technical Skills

Technologies: Python, SQL, pandas, NumPy, SciPy, Matplotlib, scikit-learn, CatBoost, LightGBM, TensorFlow, Jupyter, Git, Bash, Linux.
Languages: Spanish (native), English (B2).

Technical Projects

Customer churn prediction for Interconnect | TripleTen Bootcamp | 2026 GitHub

- **Objective:** Develop a machine learning model to identify customers at risk of canceling a telecommunications service.
- **Data:** 7,043 customers with information on contract type, monthly charges, tenure, internet services, and phone services.
- **Techniques:** Data preprocessing and exploratory data analysis, feature engineering based on EDA findings, and hyperparameter tuning with CatBoost.
- **Result:** Achieved a ROC-AUC score of 0.9253 on the test set, surpassing the project target of 0.88.
- **Technologies:** Python, pandas, NumPy, scikit-learn, CatBoost, matplotlib, and seaborn.

Experience

Data science projects | TripleTen bootcamp *Dec 2025 – May 2026*

- Developed end-to-end data science projects using Python and SQL, from preprocessing and exploratory analysis to model evaluation.
- Built and optimized machine learning models for classification, regression, time series, natural language processing, and computer vision tasks.
- Performed feature engineering, feature selection, and hyperparameter tuning to improve model predictive performance.
- Applied statistical analysis and hypothesis testing to identify patterns in data and support decision-making.
- Used Git and GitHub for version control, technical documentation, and reproducible project development.

AI model evaluator | Freelancer *Dec 2024 – Present*

Outlier.ai & Alignerr - Remote project

- Evaluate responses generated by large language models, analyzing logical reasoning, mathematical accuracy, programming, and instruction-following.
- Compare and rank responses using criteria such as factual accuracy, coherence, and quality, contributing to the training and improvement of generative AI systems.

Data analyst | MEGASTAR project *May 2023 – Dec 2024*

National Institute of Astrophysics, Optics and Electronics (INAOE) & Gran Telescopio Canarias

- Designed and implemented Python and Bash tools to automate the large-scale execution of the MEGARA Data Reduction Pipeline.
- Processed and validated 2007 high-resolution stellar spectra for inclusion in the public MEGASTAR library, exceeding the initial assigned target by more than four times.
- Implemented quality control procedures for metadata, coordinates, spectral classifications, and calibrations.
- **Impact:** Reduced processing time from approximately 30 days of manual work to 34 hours of automated execution, a reduction of nearly 95%.
- **Technologies:** Python, Bash, Linux, Git, Jupyter Notebook.

Education

Master of Science in Astrophysics *Jul 2022 – May 2024*
National Institute of Astrophysics, Optics and Electronics (INAOE)
Thesis: MEGARA Stellar Spectral Library: Sample 2

Bachelor of Science in Applied Mathematics, emphasis in Computational Sciences *Jul 2015 – Jan 2021*
Autonomous University of the State of Hidalgo